Uch o'lchamli panjarada fazoviy jismlarni chizish

Kub va kuboidni nuqtali panjarada chizishni bilamiz. Endi prizma, piramida, silindr va konusni ham shunday chizamiz: qog'ozda ular hajmli bo'lib ko'rinadi.

1-QISM

Mavzu

ESLAB OLAYLIK

- Uch o'lchamli panjarada nuqtalar uch yo'nalishda qator bo'lib turadi: **tik**, **chapga-qiya** va **o'ngga-qiya**. Ikki qo'shni nuqta orasi — **1 birlik**.
- Kub va kuboid chizmasida faqat ko'rinadigan 9 ta qirra chiziladi. Orqadagi qirralar chizilmaydi.
- Prizma va silindrning ikkita bir xil asosi bor. Piramida va konusning bitta asosi va bitta uchi bor.

USUL · KUB, KUBOID, PRIZMA VA SILINDRNI CHIZISH

1. Oldimizda ko'rinib turgan **asosni** chizamiz. Uning uchlari panjara nuqtalarida bo'ladi.
2. Asos uchlaridan **bir xil uzunlikdagi parallel chiziqlar** tortamiz. Jism orqasida qoladigan chiziq chizilmaydi.
3. Chiziqlarning uchlarini tutashtiramiz. Ikkinchi asosdan faqat ko'rinadigan qismi chiziladi.

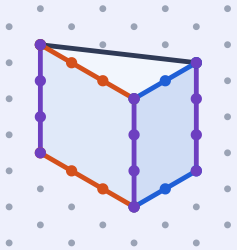
Uchburchakli prizma: asosidagi qirralar 3 va 2 birlik, balandligi 3 birlik.



1. Ustki asos



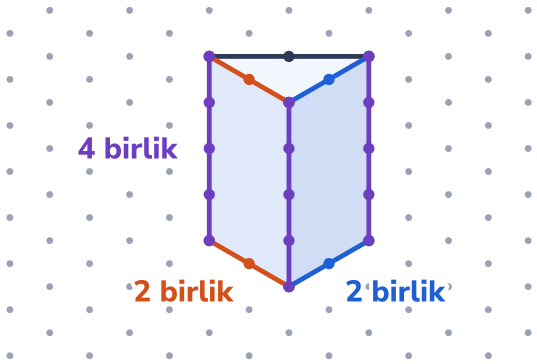
2. 3 birlik pastga



3. Pastki asos

1-MISOL · TIK TURGAN PRIZMA

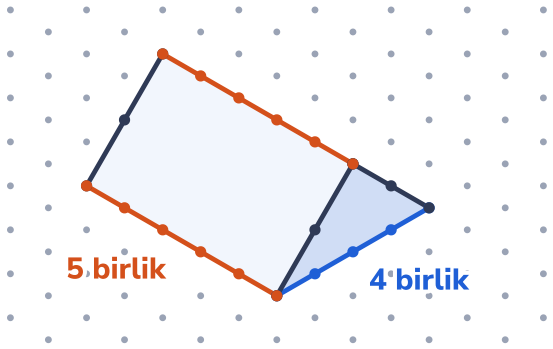
Asosi — uchburchak, uning ikki qirrasi 2 birlikdan. Balandligi 4 birlik.



1. Ustki uchburchak: chapga-qiya 2 birlik, o'ngga-qiya 2 birlik, uchlari tutashtiriladi.
2. Uchala uchidan pastga 4 birlikdan tik chiziq.
3. Pastki uchlarni tutashtiramiz: ko'rinadigani — 2 ta qirra.

2-MISOL · YOTGAN PRIZMA (TOM)

Uy tomiga o'xshagan prizma. Asosi — uchburchak: pastki qirrasi 4 birlik, uchi 2 birlik balandda. Uzunligi 5 birlik.



1. Old uchburchak: pastki qirrasi o'ngga-qiya 4 birlik, uchi uning o'rtasidan 2 birlik yuqorida.
2. Uchburchak cho'qqisidan va chap pastki uchidan chapga-qiya 5 birlikdan chiziq. O'ng pastki uchidan chizilmaydi: bu chiziq tom orqasida qoladi.
3. Uchlarini tutashtiramiz: orqa uchburchakdan bitta qirra ko'rinadi.

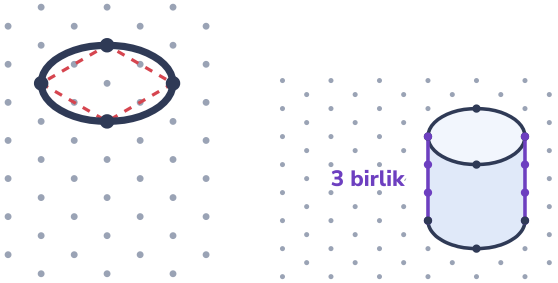
USUL · OVALNI CHIZISH

Silindr va konusning asosi — doira. Panjarada u cho'zinchoq **oval** bo'lib ko'rinadi.

Panjarada kvadrat belgilaymiz — u **romb** bo'lib ko'rinadi. Oval shu rombning **4 ta uchidan** silliq o'tadi.

3-MISOL · TIK TURGAN SILINDR

Oval — tomoni 2 birlik kvadrat orqali, balandligi 3 birlik.

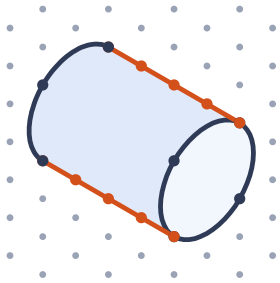


1-qadam tayyor

1. Ustki oval: rombning 4 uchi orqali.
2. Ovalning eng chap va eng o'ng nuqtasidan pastga 3 birlik.
3. Pastki ovalning faqat oldingi yarmi chiziladi.

4-MISOL · YOTGAN SILINDR (QUVUR)

Oval — tomoni 2 birlik kvadrat orqali, uzunligi 4 birlik.



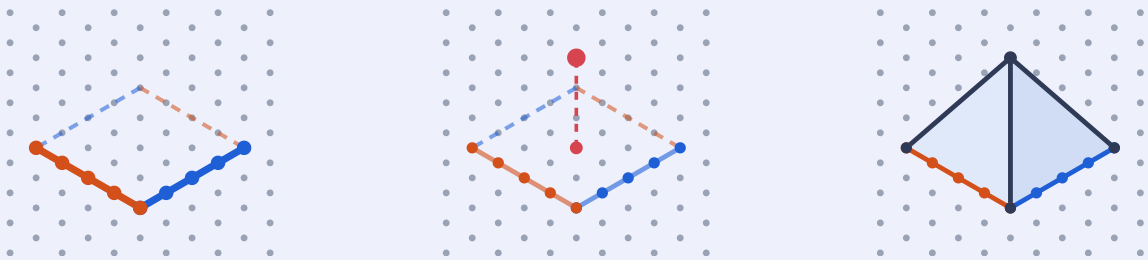
1. Old oval: romb tik turadi, oval uning 4 uchidan o'tadi.
2. Rombning eng yuqori va eng past uchidan chapga-qiya 4 birlik.
3. Orqa ovaldan faqat ko'rinadigan yarmi chiziladi.

USUL · PIRAMIDA VA KONUSNI CHIZISH

1. **Asosni** chizamiz: piramidada — ko'pburchak, konusda — oval.
2. Asos ichidagi nuqtadan (kvadrat, to'g'ri to'rtburchak va ovalda — o'rtasidan) tik yuqoriga balandlik sonicha birlik sanab, **uchini** belgilaymiz.
3. Uchini asosning ko'rinadigan uchlari bilan tutashtiramiz. Konusda uchidan ovalning ikki chetiga ikkita chiziq tortiladi.

Asosning orqa qirralarini avval ingichka chizib olsak, o'rtasini topish oson. Chizma tugagach, ular o'chiriladi.

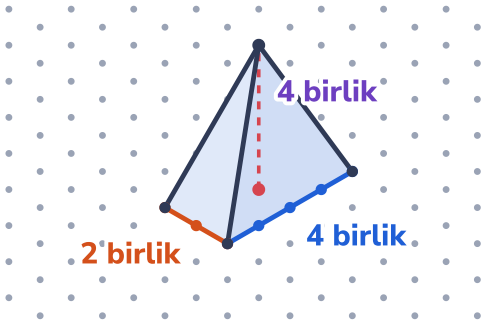
Kvadrat asosli piramida: asosi 4 × 4, balandligi 3 birlik.



1. Asos **2.** Uchi: 3 birlik yuqorida **3.** Yon qirralar

5-MISOL · TO'G'RI TO'RTBURCHAK ASOSLI PIRAMIDA

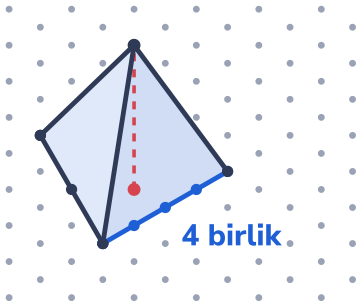
Asosi 2 × 4 birlik, balandligi 4 birlik.



1. Asos: chapga-qiya 2 birlik, o'ngga-qiya 4 birlik.
2. Asosning o'rtasi — qizil nuqta. Undan 4 birlik yuqorida — uchi.
3. Uchini asosning ko'rinadigan 3 ta uchi bilan tutashtiramiz.

6-MISOL · UCHBURCHAK ASOSLI PIRAMIDA

Asosi — uchburchak. Uchi asos ichidagi qizil nuqtadan 4 birlik yuqorida.



1. Asos — uchburchak, uchlari panjara nuqtalarida.
2. Qizil nuqtadan 4 birlik yuqoriga — uchi.
3. Uchini asosning uchala uchi bilan tutashtiramiz. Asosning orqa qirrasi ko'rinmaydi.

7-MISOL · KONUS

Oval — tomoni 2 birlik kvadrat orqali, balandligi 4 birlik.



1. Pastki oval — rombning 4 uchi orqali.
2. Oval o'rtasidan 4 birlik yuqorida — uchi.
3. Uchidan ovalning ikki chetiga chiziq. Orqa yarmi ko'rinmaydi.

8-MISOL · UCHI PASTDA TURGAN KONUS

Muzqaymoq stakanchasiga o'xshaydi: asosi tepada, uchi 3 birlik pastda.



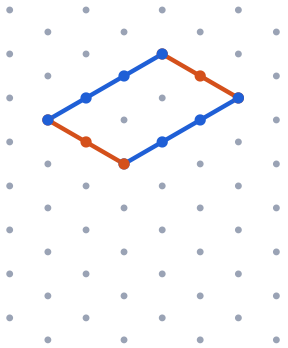
1. Ustki oval to'liq chiziladi: unga tepadan qarab turibmiz.
2. Oval o'rtasidan 3 birlik pastda — uchi.
3. Uchidan ovalning ikki chetiga chiziq.

2-QISM

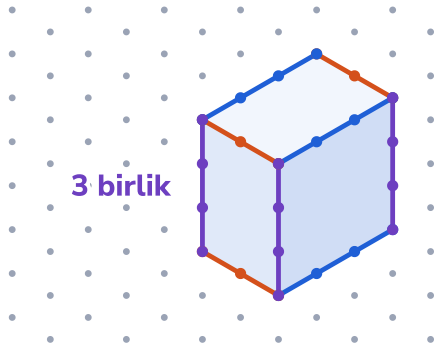
Misollar ishlash

1-MISOL · CHIZMANI YAKUNIGA YETKAZISH (KUBOID)

Kuboidning ustki yog'i chizilgan. Balandligi 3 birlik. Chizmani tugating.



berilgan



tugatilgan

1. Ustki yoqning old, chap va o'ng uchlaridan pastga 3 birlikdan tik chiziq. Eng orqadagi uchidan chiziq tushirilmaydi: u kuboid orqasida qoladi.
2. Tik chiziqlarning pastki uchlarini tutashtiramiz: 2 ta qirra.

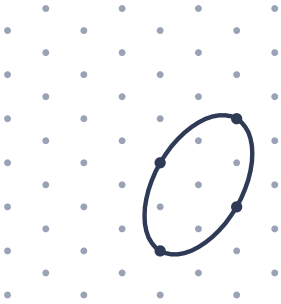
Balandligi 1 birlik bo'lganida ham ustki yoq o'zgarmas edi — faqat tik chiziqlar 1 birlik bo'lardi.



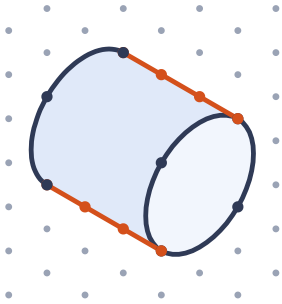
Stakanning og'ziga yon tomondan qarasam, doira emas, cho'zinchoq oval ko'rinadi. Yotgan quvurning og'zi esa qiyshiq oval bo'lib ko'rinadi.

2-MISOL · CHIZMANI YAKUNIGA YETKAZISH (YOTGAN SILINDR)

Silindrning old ovali chizilgan. Uzunligi 3 birlik. Chizmani tugating.



berilgan

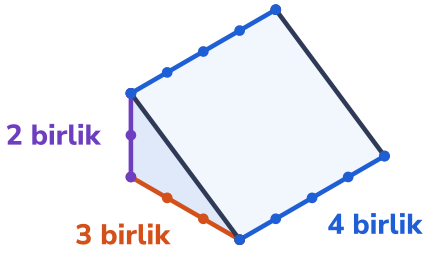


tugatilgan

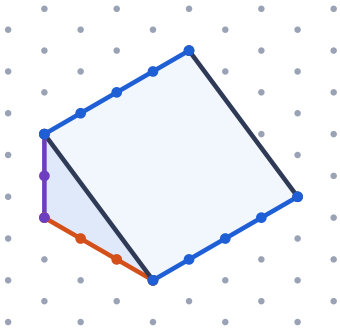
1. Ovalda 4 ta nuqta belgilangan — tik turgan rombning uchlari. Ulardan eng yuqorisi va eng pastidan chapga-qiya 3 birlikdan chiziq.
2. Ikki chiziq uchlari orasiga orqa ovalning ko'rinadigan yarmini chizamiz. U old ovalga o'xshab egiladi.

3-MISOL · RASMDAGI JISMNI PANJARADA CHIZISH

Rasmdagi prizmani uch o'lchamli panjarada chizing.



rasm



panjarada

1. Rasmda qaysi qirra qaysi yo'nalishda ekanini aniqlaymiz: 3 birlik — chapga-qiya, 2 birlik — tik, 4 birlik — o'ngga-qiya.
2. Old uchburchak: chapga-qiya 3 birlik va tik 2 birlik, uchlarini tutashtiramiz.
3. Uchburchakning tepa uchidan va o'ng pastki uchidan o'ngga-qiya 4 birlikdan chiziq (chap pastki uchidan chiqadigan chiziq orqada qoladi), so'ng ularning uchlarini tutashtiramiz.

Uzunligi 4 emas, 6 birlik bo'lsa — faqat o'ngga-qiya chiziqlar uzayadi, uchburchak o'sha-o'sha.

Chizib bo'lgach tekshiraman: parallel chiziqlar bir xil uzunlikdami, orqadagi chiziqlar qolib ketmadimi?



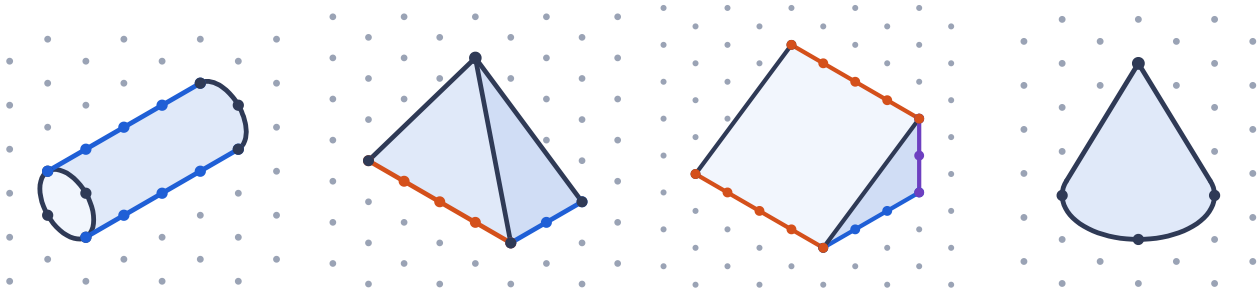
DIQQAT!

1. Orqadagi chiziqlar chizilmaydi: ✗ kuboidning 12 ta qirrasi chizilgan ✓ **faqat ko'rinadigan 9 ta qirra**
2. Silindrning pastki asosi: ✗ butun oval chizilgan ✓ **faqat oldingi yarmi**
3. Birlik — nuqtalar orasidagi oraliq: ✗ 3 birlik uchun 3 ta nuqta ✓ **3 birlik = 3 oraliq = 4 ta nuqta**
4. Piramida uchi: ✗ asosning bir uchi ustida ✓ **asos ichidagi nuqta ustida (kvadrat asosda — o'rtasi ustida)**
5. Prizmaning parallel chiziqlari: ✗ har xil uzunlikda ✓ **hammasi bir xil uzunlikda**

3-QISM

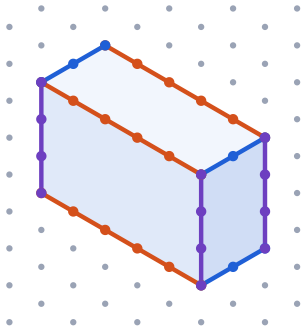
Mustahkamlash

1. Chizmaga qarab jismning nomini ayting.

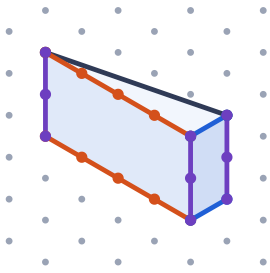


a) b) c) d)

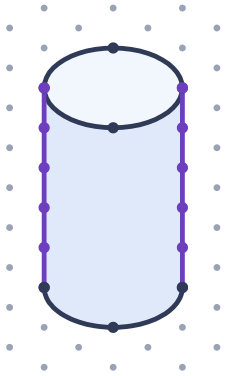
2. Chizmadagi jismning o'lchamlarini toping (oraliqlarni sanang).



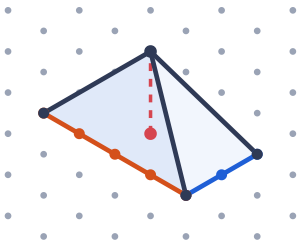
a) bo'yi, eni, balandligi



b) asos qirralari, balandligi



c) balandligi



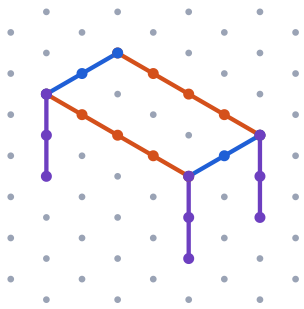
d) asosi, balandligi

3. a) Kub chizmasida nechta qirra ko'rinadi?
   b) Tik turgan silindr chizmasida nechta to'g'ri chiziq bor?
   c) 1-qismdagi kvadrat asosli piramida chizmasida nechta chiziq bor?
   d) Uzunligi 5 birlik bo'lgan tik chiziq uchlari bilan nechta nuqtadan o'tadi?

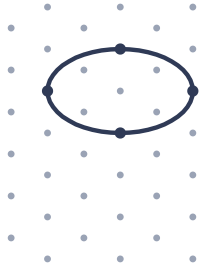
4. Buyum qaysi jismga o'xshaydi?
   a) sharbat qutisi; b) yo'lga qo'yiladigan to'siq konusi; c) suv quvuri; d) chordoqli uyning tomi; e) muzqaymoq stakanchasi.

5. Daftaringizdagi nuqtali panjarada chizing.
   a) Qirrasi 4 birlik bo'lgan kub.
   b) Bo'yi 2, eni 5, balandligi 1 birlik bo'lgan kuboid.
   c) Tik turgan prizma: asosi uchburchak, asosidagi ikki qirrasi 1 va 4 birlik, balandligi 3 birlik.
   d) Piramida: asosi 4×4 kvadrat, balandligi 5 birlik.
   e) Silindr: asosi tomoni 2 birlik bo'lgan kvadrat orqali chizilgan oval, balandligi 2 birlik.
   f) Konus: asosi tomoni 2 birlik bo'lgan kvadrat orqali chizilgan oval, balandligi 5 birlik.

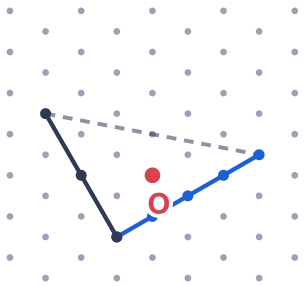
6. Chizmalarni daftaringizga ko'chirib, yakuniga yetkazing.



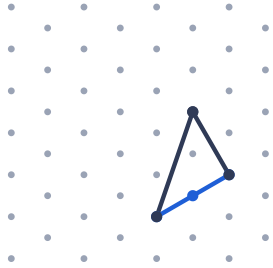
a) kuboid



b) silindr, balandligi 2

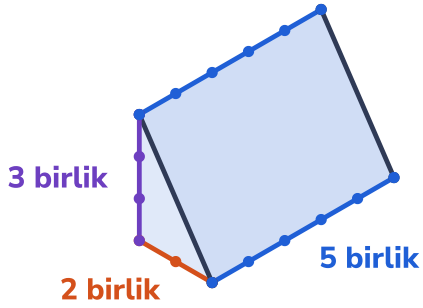


c) piramida, uchi O dan 3 birlik yuqorida

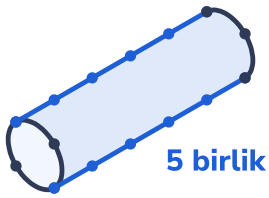


d) prizma, uzunligi 3

7. Rasmdagi jismlarni nuqtali panjarada chizing.

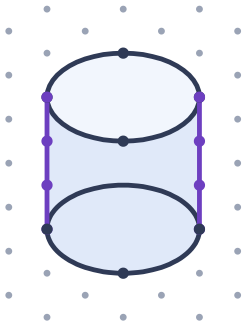


a)



b) asosi tomoni 1 birlik kvadrat orqali

8. **Xatoni toping.** Sardor tik turgan silindrni chizdi (rasmda). Uning chizmasida nima ortiqcha?



9. Prizma chizmasi 2-qadamgacha chizilgan (rasmda). Chizmani tugatish uchun yana nechta chiziq kerak? Ular qaysi yo'nalishda?



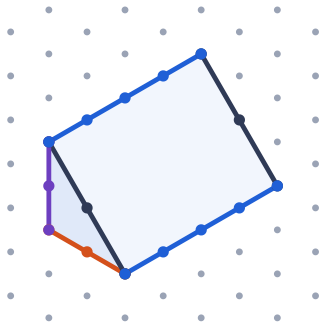
10. Silindr va konusning asosi bir xil oval, balandligi ham bir xil — 3 birlik. Ularning chizmalari nimasi bilan farq qiladi?

11. **Qiyinroq.** Qirrasi 3 birlik bo'lgan kub va bo'yi 2, eni 3, balandligi 4 birlik bo'lgan kuboid chizildi. Qaysi chizmadagi chiziqlarning umumiy uzunligi katta?

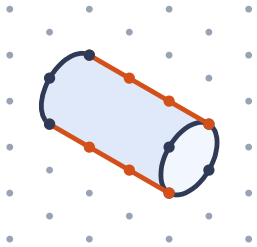
4-QISM

Mustaqil ish

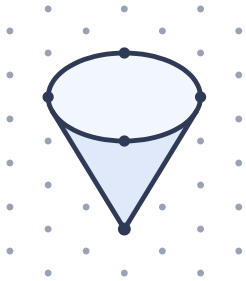
1. Jismning nomini va asosi qanday shakl ekanini yozing.



a)

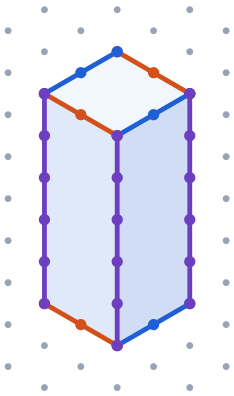


b)

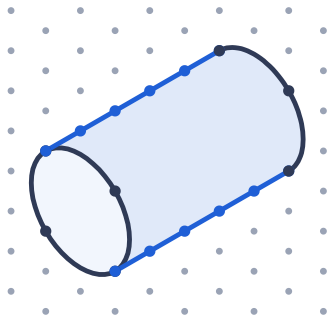


c)

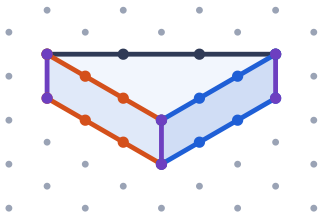
2. Chizmadagi jismning o'lchamlarini toping.



a) bo'yi, eni, balandligi



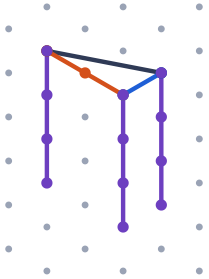
b) uzunligi



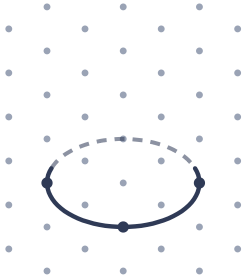
c) asos qirralari, balandligi

3. Nuqtali panjarada chizing.
   a) Bo'yi 4, eni 3, balandligi 2 birlik bo'lgan kuboid.
   b) Tom shaklidagi yotgan prizma: asosining pastki qirrasi 2 birlik, uchi 1 birlik balandda, uzunligi 6 birlik.
   c) Silindr: asosi tomoni 1 birlik bo'lgan kvadrat orqali chizilgan oval, balandligi 6 birlik.
   d) Konus: asosi tomoni 2 birlik bo'lgan kvadrat orqali chizilgan oval, balandligi 2 birlik.

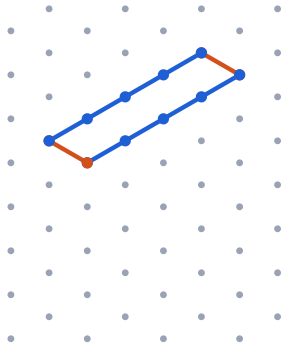
4. Chizmalarni ko'chirib, yakuniga yetkazing.



a) prizma

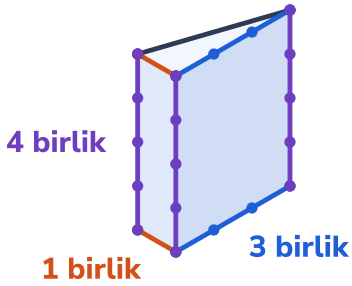


b) konus, balandligi 3

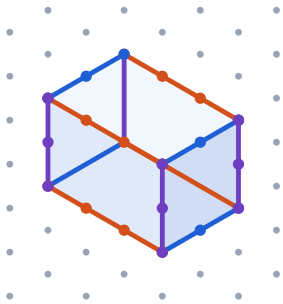


c) kuboid, balandligi 3

5. Rasmdagi prizmani nuqtali panjarada chizing.



6. **Xatoni toping.** Malika kuboidni chizdi (rasmda). Uning chizmasida nima ortiqcha?



7. **Qiyinroq.** Qirrasi 4 birlik bo'lgan kub chizmasidagi chiziqlarning umumiy uzunligi bo'yi 5, eni 3 birlik bo'lgan kuboid chizmasidagi chiziqlarnikiga teng. Kuboidning balandligi necha birlik?

8. Uyingizdan prizma yoki silindr shaklidagi ikkita buyum toping. Ularni nuqtali panjarada chizing.

5-QISM

Javoblar

Mustahkamlash

1-masala

a) silindr (yotgan); b) piramida (asosi to'g'ri to'rtburchak); c) prizma (asosi uchburchak); d) konus.

2-masala

a) bo'yi 5, eni 2, balandligi 3. b) asos qirralari 4 va 1, balandligi 2. c) balandligi 5. d) asosi 4×2, balandligi 2.

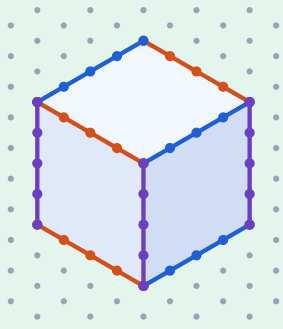
3-masala

a) 9 ta. b) 2 ta. c) 5 ta: asosning 2 ta qirrasi va 3 ta yon qirra. d) 5 oraliq — $5 + 1 = 6$ ta nuqta.

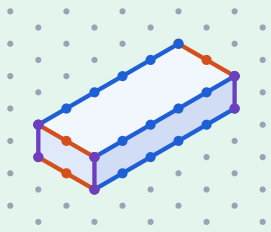
4-masala

a) kuboid; b) konus; c) silindr; d) prizma; e) konus (uchi pastda).

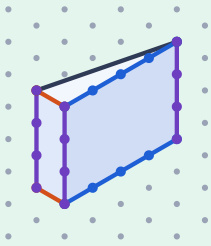
5-masala



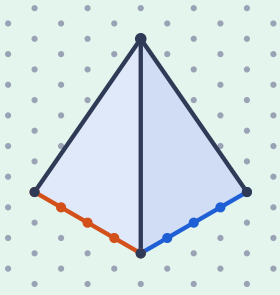
a)



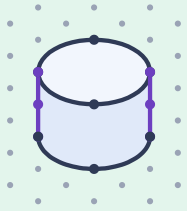
b)



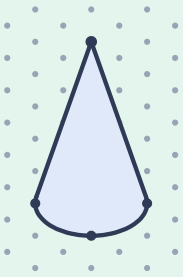
c)



d)

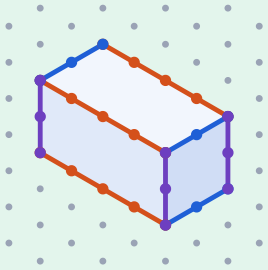


e)

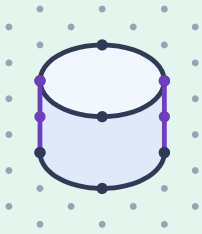


f)

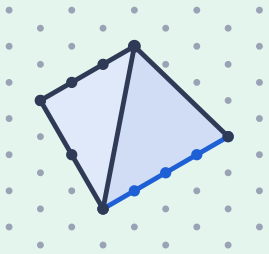
6-masala



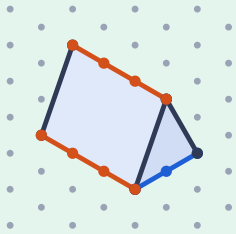
a) pastki 2 ta qirra



b) 2 ta tik chiziq va yarim oval

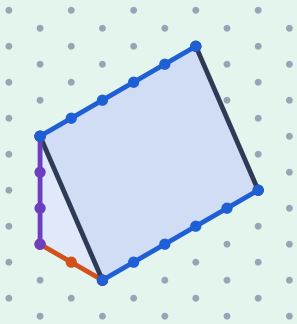


c) 3 ta yon qirra

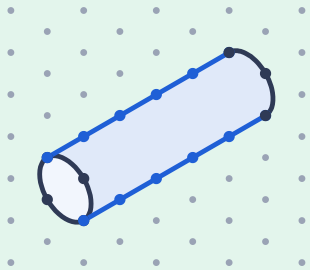


d) 2 ta parallel chiziq va orqa qirra

7-masala



a)



b)

a) Old uchburchak: chapga-qiya 2 birlik va tik 3 birlik; tepa va o'ng pastki uchidan o'ngga-qiya 5 birlikdan chiziq. b) Old oval — bitta katak rombining 4 uchi orqali; ikki chetidan o'ngga-qiya 5 birlikdan chiziq.

8-masala

Pastki ovalning orqa yarmi ortiqcha: u silindrning orqasida qoladi va ko'rinmaydi. Pastki asosdan faqat oldingi yarmi chiziladi.

9-masala

Yana 2 ta chiziq — pastki uchburchakning ko'rinadigan qirralari: biri chapga-qiya (3 birlik), biri o'ngga-qiya (3 birlik).

10-masala

Silindrda ustki oval to'liq, pastki ovalning yarmi va ikkita tik chiziq bor. Konusda bitta ovalning oldingi yarmi bor, ikkita qiya chiziq esa uchida — bitta nuqtada uchrashadi.

11-masala

Ikkala chizmada ham 9 ta qirra. Kub: $9 \cdot 3 = 27$ birlik. Kuboidda har yo'nalishda 3 tadan qirra: $3 \cdot 2 + 3 \cdot 3 + 3 \cdot 4 = 6 + 9 + 12 = 27$ birlik. Uzunliklari teng.

Mustaqil ish

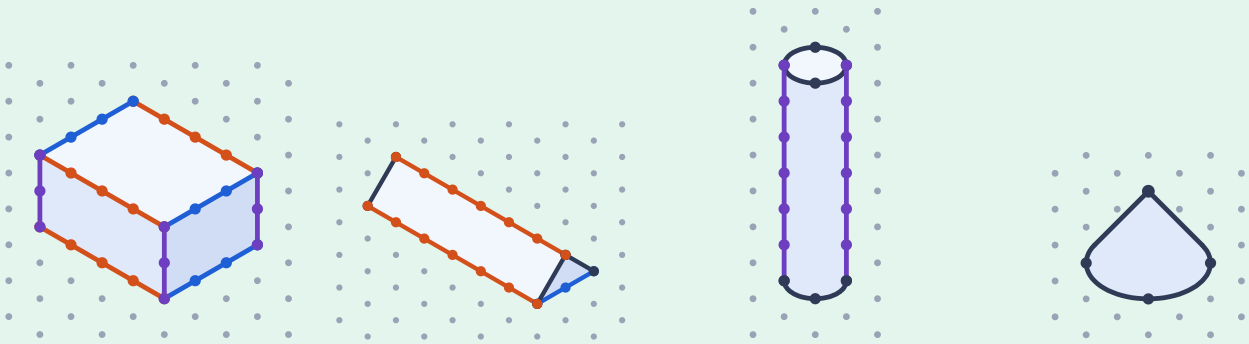
1-masala

a) prizma, asosi uchburchak; b) silindr, asosi doira (panjarada oval); c) konus, asosi doira (uchi pastda).

2-masala

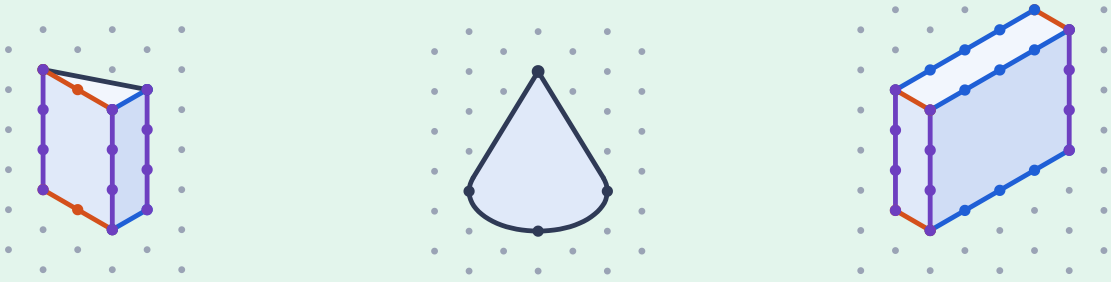
a) bo'yi 2, eni 2, balandligi 5. b) uzunligi 5. c) asos qirralari 3 va 3, balandligi 1.

3-masala



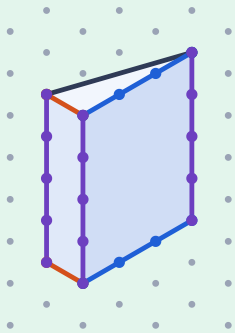
a) b) c) d)

4-masala



a) pastki 2 ta qirra b) uchi va 2 ta chiziq c) 3 ta tik va 2 ta pastki qirra

5-masala



Ustki uchburchak: chapga-qiya 1 birlik va o'ngga-qiya 3 birlik; uchidan pastga 4 birlikdan 3 ta tik chiziq; pastda 2 ta qirra.

6-masala

Orqadagi 3 ta qirra ortiqcha — ular ko'rinmaydi. To'g'ri chizmada 9 ta qirra bo'ladi.

7-masala

Kub: $9 \cdot 4 = 36$ birlik. Kuboidda har yo'nalishda 3 tadan qirra: $3 \cdot 5 + 3 \cdot 3 = 24$, qolgani $36 - 24 = 12$ — bu 3 ta tik qirra. Balandligi $12 : 3 = 4$ birlik.

8-masala

Masalan: tish pastasi qutisi — kuboid (prizma), choy qutisi — silindr. Chizishda 1-qismdagi usuldan foydalaning.